ABSTRACT

Project title:- Digital Twins Using Artificial Intelligence

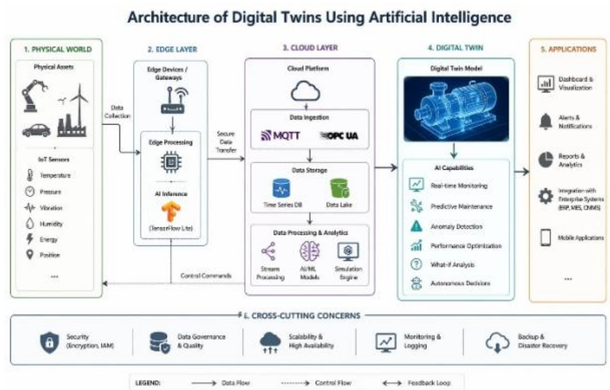
Objective:- The primary objective of developing a Digital Twin with Artificial Intelligence (DTAI) and structured data to create dynamic, self-learning ,copy of an object of real world process, or system. Some of the important Objectives are real-time monitoring product design improvement, performance optimisation ,Life cycle management .At last it help in monitoring overall performance ,predicting, upgrading real-world systems efficiently.

***Introduction*:-** Digital twins are dynamic,non real representation of a physical asset ,system or that procedure that remains time to time updated using real time data of IOT sensors. When we add AI for virtual models they can learn their own and real world they no longer exist as real objects They become in such a way that the can learn there own and guess what will happen in the future and take action by themselves without human interaction.

Technical stacks:- digital twins are built by taking the help of sensors, edge devices ,cloud .Sensors are used to sense and collect data from real life applications. Tensor flow lite is used to detect problems on devices. tools like MQTT and OPC-UA help in communication .it explains how smart technologies work faster automatically.

Methodology :- Digital twin is a duplicate copy of a object .it stays connected to real object and update time to time .sensors used to transfer datafrom real life applications to the digital model. It is used in different situations virtually. The system can also improve performance automatically.and the data which is stored and organised way for easy to use. In a line it is a smart mode that tracks and helps to improve real world system applications automatically.

Architecture:-



Conclusion:- Digital twins using artificial intelligence is used to create smart digital copy or duplicate of a real life applications of the world that can check each information and caution and improve it and implement in the real world application they are largely used in healthcare, industries like manufacturing and exporting and smart cities data structure. It is used for better data analysing and making better decision .they are completely turned into self learning systems when combined with AI. They all together create efficient and automated systems.

Future Enhancement :-

1.Better in storing and organising and managing the data
2.Integrating with future technologies.
3. Advanced and secured in protection.
4. Smart and sustainable resource optimization
5. System or platform integration
6. Managing large systems efficiently

Name :- V.sarthak

Branch&section :- CSE-F

Roll number:- 25P61A05Z1